# Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

**Problem B**

**Abstract**

Gravity, drag and Magnus lift govern a soccer ball's three-dimensional flight. We formulate a constant-spin model and compare gravity alone, constant drag, constant drag and lift, and complete variable coefficients. Empirical aerodynamic relations and a matched wind-tunnel comparison provide the coefficient basis.

Five prescribed kick situations give six initial states and 24 forward flights, illustrating the roles of speed, elevation and spin direction. Three actual match penalties provide 39 original sampled frames. After conversion to penalty-mark coordinates, three selected positions per event enter 12 bounded inversions, whose continuations are compared with later measured samples. Per-penalty tables give measured fit positions, inverse initial states, predicted crossings and crossing deviations, with termination states marked. A separate same-state curved-shot calculation quantifies angular damping, shifting the crossing by 0.0343 m. Section 5 uses the competition positions as a reference example for individual parameter effects, crossings compatible with the same early positions, and 100,000 Monte Carlo trajectories with fixed mass and radius.

**Keywords:** Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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# 1 Introduction

## 1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [10]; Maponi and Piombin developed a numerical framework that also accommodates wind [8].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [11], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [9]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [15]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [6, 7], allowing predicted crossings to be compared with later recorded positions.

## 1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions determine the forward examples and the unknown quantities reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 examines these sensitivities and joint perturbations.

We first compare prescribed kicks, then estimate initial states from three real penalties and compare predictions with later recorded positions. Figure 1 summarizes this physical and computational sequence.

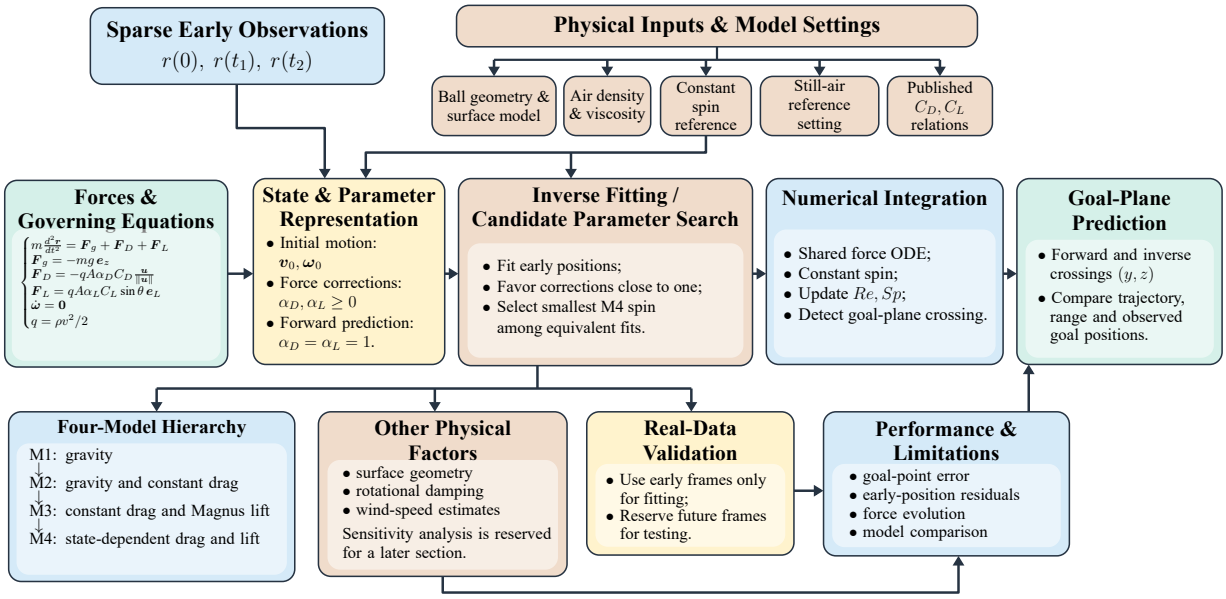



Figure 1: Three-force modeling, prescribed forward scenarios and real-penalty inverse comparison.

# 2 Preparation of the Models

## 2.1 Assumptions

- **Contact and model start.** Prescribed examples begin just after contact. Real inversions define $t = 0$ at the selected starting tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [8, 5, 10]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [8, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.

- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.
- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 uses the competition-position example to examine parameter effects on crossing distributions.

## 2.2 Notations

Table 1: Principal quantities and competition reference values.

| Symbol | Meaning | Value or unit |
|---|---|---|
| $\boldsymbol{r}, \boldsymbol{v}$ | Ball-centre position and velocity | m; m/s |
| $\boldsymbol{\omega}, \omega$ | Spin vector and its magnitude | rad/s |
| $m, R, A$ | Mass, radius and frontal area | 0.43 kg; 0.11 m; $\pi R^2$ |
| $g$ | Gravitational acceleration | 9.81 m/s$^2$ |
| $\boldsymbol{u}, v$ | Air-relative velocity and its magnitude | m/s |
| $\rho$ | Air density | 1.2 kg/m$^3$ |
| $\nu, \mu = \rho\nu$ | Kinematic and dynamic viscosity | $1.50 \times 10^{-5}$ m$^2$/s; Pa s |
| $Re, Sp$ | Reynolds number and spin parameter | dimensionless |
| $C_D, C_L$ | Drag and transverse lift coefficients | dimensionless |
| $\alpha_D, \alpha_L$ | Multiplicative force corrections | dimensionless |
| $\theta, \boldsymbol{e}_L$ | Spin–airflow angle and Magnus direction | rad; unit vector |
| $X_g$ | Local goal-plane coordinate | 11 m |
| $\boldsymbol{Q}, I_b$ | Damping torque and moment of inertia (Section 6) | N m; kg m$^2$ |

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [8]; the air density follows Asai et al. [11]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

# 3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

## 3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with $x$ towards the goal, $y$ lateral and $z$ upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.2 converts the measured pitch-centred coordinates.

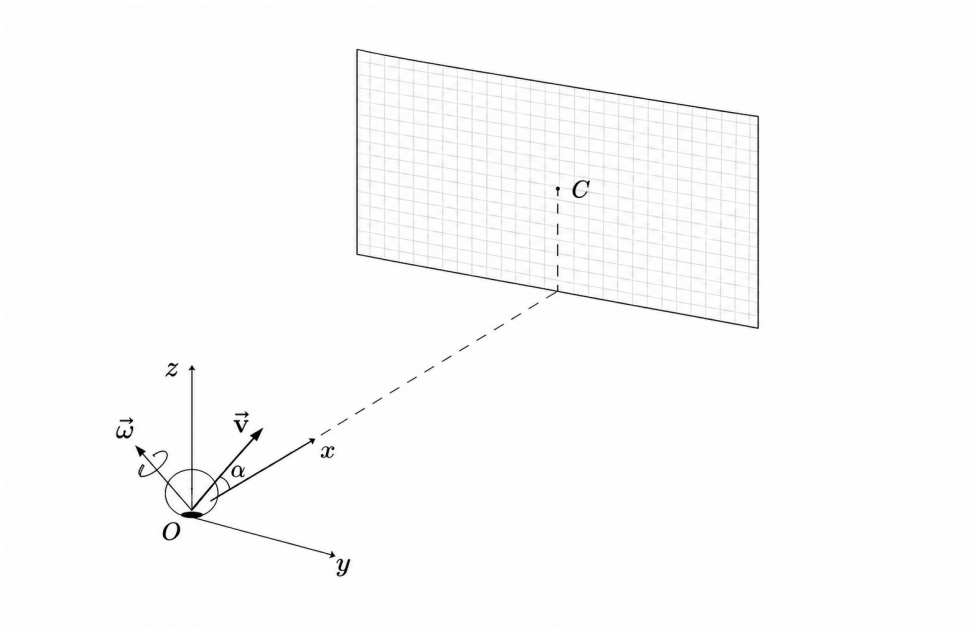



Figure 2: Penalty-kick geometry in local coordinates; $O$ is the nominal penalty mark and $C$ an illustrative goal-plane point.

A clean opening crossing requires $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

## 3.2 Force Analysis

### 3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

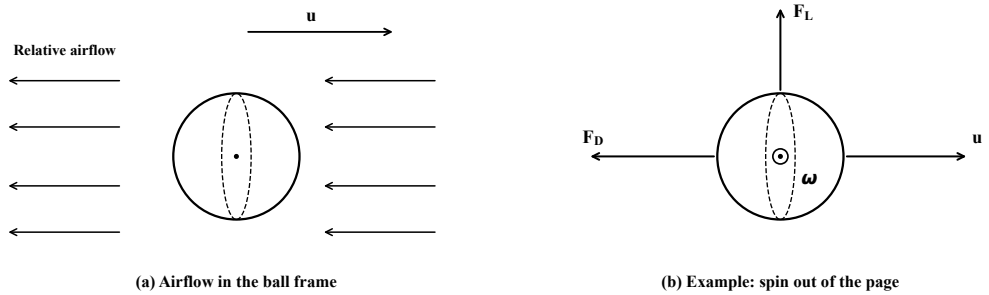
By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about $1.56\%$ of the ball's weight and changes the downward acceleration from $g$ to $g(1 - \rho V_b/m)$.

The 1.56% buoyancy scale is a small vertical correction; all four main comparisons use gravity, drag and Magnus lift as defined above.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.




(a) Airflow in the ball frame (b) Example: spin out of the page

Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

### 3.2.2 Aerodynamic Drag

In the ball frame, the far-field air velocity is $-\boldsymbol{u}$. Approximate the maximum cross-section by a disk of area $A = \pi R^2$ facing the oncoming air. For a uniform stream decelerated from speed $v$ to rest, Bernoulli's relation gives the stagnation-to-far-field pressure difference and a corresponding force scale,

$$\Delta p_{\text{stag}} = \tfrac{1}{2}\rho v^2, \qquad F_{\text{scale}} = \Delta p_{\text{stag}} A = \tfrac{1}{2}\rho A v^2. \tag{3}$$

Pressure and viscous shear vary across the actual ball, so the dimensionless coefficient $C_D$ expresses the integrated drag relative to this scale. Drag opposes the velocity of the ball relative to the air, giving [15, 12]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A \alpha_D C_D(Re, Sp) v \boldsymbol{u}, \qquad \boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}, \quad v = \|\boldsymbol{u}\|. \tag{4}$$

Its quantities have the following physical roles:

1. **Air density $\boldsymbol{\rho}$.** The density measures air mass per unit volume. We use the uniform reference value $1.2$ kg/m$^3$ reported by Asai et al. [11].

2. **Projected area $\boldsymbol{A = \pi R^2}$.** A larger radius intercepts more airflow; the spherical geometry fixes this area throughout flight.

3. **Relative speed and direction $\boldsymbol{v, u}$.** The quadratic speed dependence gives stronger resistance at higher speed, and the negative direction opposes airflow. The still-air main model has $\boldsymbol{u} = \boldsymbol{v}$.

4. **Drag coefficient $\boldsymbol{C_D(Re, Sp)}$.** The coefficient represents boundary-layer transition, separation and wake formation. Following Asai et al. and Kiratidis and Leinweber [11, 12], we

define

$$Re = \frac{2Rv}{\nu}, \qquad Sp = \frac{R\omega}{v}, \qquad \omega = \|\boldsymbol{\omega}\|, \quad \nu = \frac{\mu}{\rho}. \tag{5}$$

$Re$ compares inertial and viscous flow effects; $Sp$ compares rotating surface speed with airspeed. Constant spin makes $Sp$ rise as the ball slows.

The drag crisis is the rapid reduction of drag associated with boundary-layer transition. A useful coefficient law should describe this transition as well as the slower variation in the supercritical regime. According to Kiratidis and Leinweber [12], their Eq. (5) represents zero-spin drag by

$$\begin{aligned} f(v) = {} & \frac{a - b_{\text{min}}}{1 + \exp[(v - v_c)/v_s]} + b_{\text{min}} \\ & + \frac{v - v_{\text{min}}}{1 + \exp[-(v - v_{\text{min}})/v_s]} \frac{b_{\text{max}} - b_{\text{min}}}{v_{\text{max}} - v_{\text{min}}}. \end{aligned} \tag{6}$$

The coefficient $a$ sets the low-speed level, $v_c$ locates the transition, and $v_s$ controls its width. The remaining parameters set the level and slope of the high-speed tail. Table 2 gives the Teamgeist parameters reported in their Table 1 [12].

Table 2: Teamgeist zero-spin drag parameters from Kiratidis and Leinweber [12].

| $a$ | $v_c$ | $v_s$ | $b_{\text{min}}$ | $b_{\text{max}}$ | $v_{\text{min}}$ | $v_{\text{max}}$ |
|---|---|---|---|---|---|---|
| — | m/s | m/s | — | — | m/s | m/s |
| 0.4927 | 12.58 | 1.071 | 0.1440 | 0.1540 | 23.17 | 35.00 |

For a spinning Teamgeist, Kiratidis and Leinweber's Eq. (7) adds a linear spin correction with slope 0.514, fitted to the supercritical wind-tunnel data [12]:

$$C_D(Re, Sp) = f\left(\frac{Re\nu}{2R}\right) + 0.514Sp. \tag{7}$$

Thus drag changes with speed even at zero spin, and also increases with rotation. The Teamgeist parameter set is used throughout the coefficient comparison; other ball surfaces require their own parameters.

The prescribed launch speeds are 14–27 m/s, giving $Re = (2.05\text{–}3.96) \times 10^5$ within the speed range represented by the Teamgeist fit. The relation describes transition and supercritical flow up to the reference 35 m/s [12]. The comparison figures below use measured $Re = 455172$; inverse flights update the coefficients at their actual $Re$ and $Sp$.

5. **Drag correction $\alpha_D \geq 0$.** This dimensionless multiplier equals one in forward calculations and is estimated from early positions in inverse calculations.

**Comparison with published wind-tunnel data.** Asai et al. [11] measured soccer-ball forces in Tokai University's low-speed recirculating wind tunnel using a six-component balance ($1.5 \times 1$ m test section; about 1% turbulence). We use their spinning full-size Teamgeist FRP replica: $R = 0.11$ m, $\rho = 1.2$ kg/m$^3$, airspeed 22–30 m/s and spin 2–10 rev/s. Five points per coefficient are digitized

from their Figures 8 and 9 at $Re = 455172$ (about 30 m/s); the reported side force $C_S$ represents transverse Magnus lift here. The speed–Re correspondence gives $\nu \simeq 1.45 \times 10^{-5}$ m$^2$/s for this matched comparison.

Figure 4 reproduces the increase of drag with $Sp$: the coefficient MAE is 0.00985 and RMSE is 0.01127. Both curves use the same experimental condition and measured spin interval.

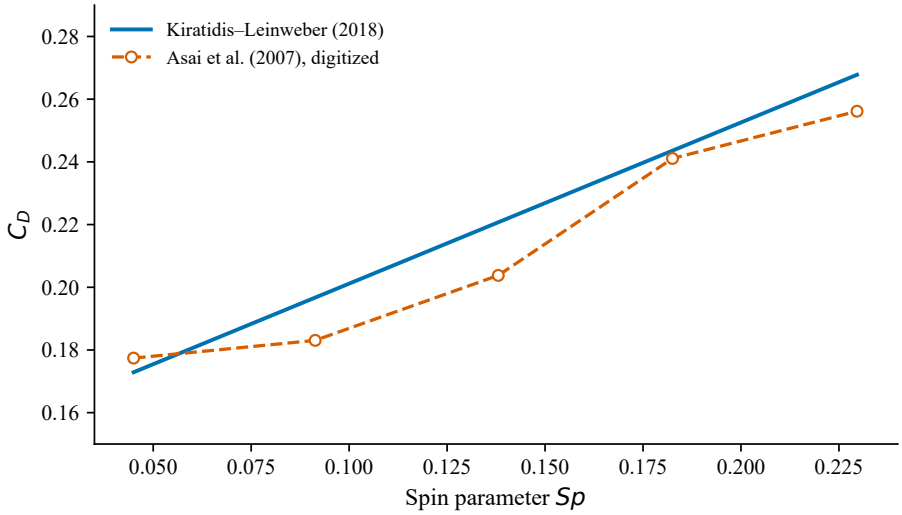



Figure 4: Teamgeist drag at $Re = 455172$: the complete empirical relation [12] and five digitized Asai measurements [11], Figure 8.

### 3.2.3 Magnus Lift

Figure 5 illustrates relative streamlines in the ball frame. For the spin shown, the upper surface moves with the relative airflow while the lower surface moves against it. Rotation makes the flow asymmetric and produces a transverse force. The figure uses a two-dimensional ideal flow with circulation to illustrate the directions.

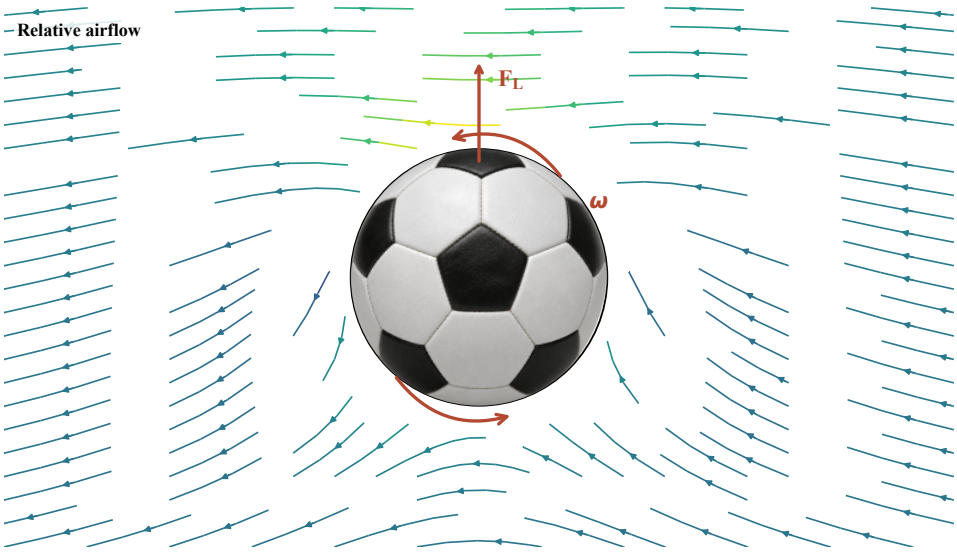



Figure 5: Idealized relative streamlines around a spinning football. Arrows show the directions of relative airflow, spin and Magnus force; colours indicate relative speed in this illustrative flow field.

Rotation changes the flow on opposite sides of the ball and creates a transverse pressure force. Using the drag dynamic-pressure scale gives $F_L = qAC_L$ for transverse spin [15, 12]. For a three-dimensional spin axis, we adopt

$$\begin{aligned} \boldsymbol{F}_L &= \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) v^2 \sin\theta \, \boldsymbol{e}_L, \\ \boldsymbol{e}_L &= \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}, \qquad \sin\theta = \frac{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}{\omega v}. \end{aligned} \tag{8}$$

Lift is zero for zero spin, zero airspeed or parallel spin and airflow.

For $\omega > 0$ and $v > 0$, substituting the definitions of direction and angle into Eq. (8) yields the equivalent form

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) \frac{v}{\omega} (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{9}$$

This form makes the direction perpendicular to both spin and relative velocity explicit. Since $C_L$ also depends on spin, the factor $1/\omega$ does not imply divergent lift at zero spin; zero spin or zero relative speed is handled separately by setting lift to zero.

1. **Dynamic-pressure and area scale $\boldsymbol{\rho A v^2/2}$.** Density, projected area and speed have the same meanings as in the drag force.

2. **Magnus direction $\boldsymbol{e_L}$.** This unit vector is perpendicular to airflow. Topspin bends the path downward, backspin upward, and sidespin horizontally.

3. **Angular projection $\boldsymbol{\sin\theta}$.** The transverse component is one for perpendicular spin and airflow and vanishes for parallel directions. This projection extends the transverse empirical law to an oblique spin axis.

4. **Lift coefficient $C_L(Re, Sp)$.** Rotation and boundary-layer state jointly determine lift.

   Rotation creates an asymmetric flow and a transverse Magnus force. Its magnitude depends on the boundary-layer state as well as spin, so a spin-only relation does not describe all flow conditions. Kiratidis and Leinweber [12] introduce the reference power law in their Eq. (8),

   $$C_L^{\text{fit}}(Sp) = 1.15 Sp^{0.83}, \tag{10}$$

   and scale it using drag as an indicator of flow state in their Eq. (9) [12]. We use that complete relation:

   $$C_L(Re, Sp) = C_L^{\text{fit}}(Sp) \frac{C_{D,\text{ceil}} - \min[C_D(Re, Sp), C_{D,\text{ceil}}]}{C_{D,\text{ceil}} - C_{D,\text{ref}}}, \tag{11}$$

   with the Teamgeist reference quantities

   $$C_{D,\text{ceil}} = f(0), \qquad C_{D,\text{ref}} = C_D(333793, 0.19). \tag{12}$$

   Here $C_{D,\text{ceil}} \simeq 0.4927$ is the zero-spin low-speed ceiling. The reference drag is evaluated with Eq. (7), including its spin term and using the same $R$ and $\nu$ as the calculation. At the reference $(Re, Sp) = (333793, 0.19)$ the scaling factor is one and $C_L \simeq 0.290$. The ceiling, reference condition and use of drag to reduce lift follow Kiratidis and Leinweber [12]; Eq. (11) supplies the final lift coefficient in the force model.

5. **Lift correction $\alpha_L \geq 0$.** This dimensionless multiplier is one in forward prediction and is fitted to early positions when inverse prediction includes lift.

**Comparison with published wind-tunnel data.** Figure 6 uses the same Asai experiment and flow condition. The complete relation reproduces increasing lift with $Sp$ and lies above all five measured values: MAE 0.04505, RMSE 0.05040 and maximum difference 0.07325. This larger discrepancy motivates the inverse lift correction $\alpha_L$.

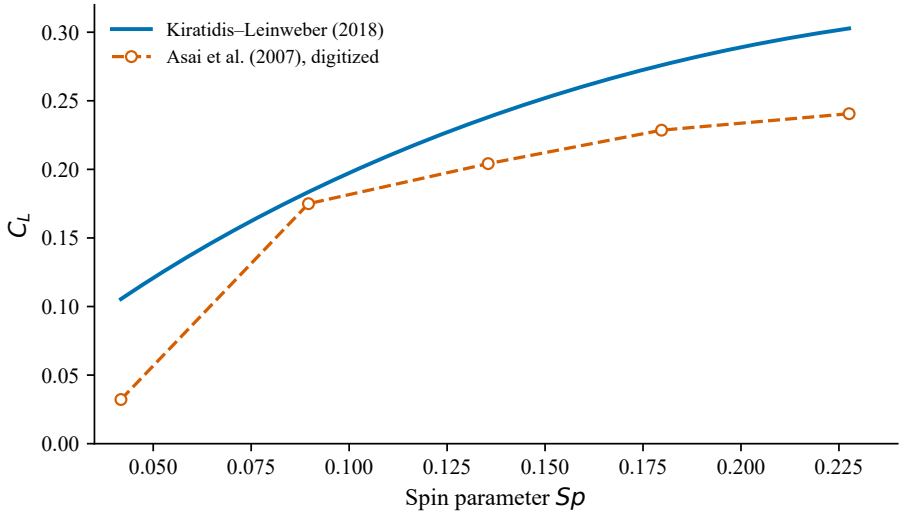



Figure 6: Teamgeist transverse lift at $Re = 455172$: the complete empirical relation [12] and five digitized Asai side-force measurements [11], Figure 9.

### 3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{13}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{14}$$

Substituting drag and Eq. (9) into Newton's second law gives the components used for numerical integration. Define $k_D = \rho A \alpha_D C_D v/(2m)$ and $k_L = \rho A \alpha_L C_L v/(2m\omega)$. For $\omega > 0$ and $v > 0$,

$$\begin{aligned} \dot{v}_x &= -k_D u_x + k_L(\omega_y u_z - \omega_z u_y), \\ \dot{v}_y &= -k_D u_y + k_L(\omega_z u_x - \omega_x u_z), \\ \dot{v}_z &= -g - k_D u_z + k_L(\omega_x u_y - \omega_y u_x). \end{aligned} \tag{15}$$

The components are integrated together with $\dot{\boldsymbol{r}} = \boldsymbol{v}$; at zero spin or relative speed the lift components are set to zero.

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{16}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

## 4 Model Solving

Forward prediction explains how a prescribed kick develops under different force assumptions. Inverse prediction then uses early positions from three actual penalties to estimate motion and predict the continuation. Both calculations use Eq. (14) and the same four-model hierarchy.

Table 3: Four-model hierarchy and actual inverse quantities. All models include gravity and constant spin.

| Model | Drag | Magnus lift | Inverse quantities |
|---|---|---|---|
| M1 | none | none | $\boldsymbol{v}_0$ (3) |
| M2 | constant $C_D^*$ | none | $\boldsymbol{v}_0, \alpha_D$ (4) |
| M3 | constant $C_D^*$ | constant $C_L^*$ | $\boldsymbol{v}_0$, spin axis, $\alpha_D, \alpha_L$ (7) |
| M4 | $C_D(Re, Sp)$ | $C_L(Re, Sp)$ | $\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L$ (8) |

The constant coefficients use the published reference point:

$$C_D^* = C_D(333793, 0.19), \qquad C_L^* = C_L(333793, 0.19) = 1.15(0.19)^{0.83}. \tag{17}$$

For the reference ball and air, $C_D^* = 0.2415$ and $C_L^* = 0.2898$. M3 uses a unit vector to represent spin direction; its constant coefficient makes the spin magnitude absent from the force. M4 uses the full spin vector.

## 4.1 Forward Prediction

### 4.1.1 Prescribed conditions and solution

Five kick situations are represented by six initial states in Table 4. These are chosen numerical examples: power, placement, sidespin, a paired topspin/backspin comparison, and a chip. Each begins at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m with

$$\boldsymbol{v}_0 = v_0(\cos\beta, 0, \sin\beta), \qquad \alpha_D = \alpha_L = 1. \tag{18}$$

The top- and backspin pair has the same speed and launch angle, isolating the direction of rotation. For every state, all four models are integrated by adaptive Runge–Kutta (DOP853). A forward root of $x - 11$ locates the goal crossing and a descending root of $z - R$ locates ground contact. The relative and absolute integration tolerances are $10^{-10}$ and $10^{-12}$, with a maximum step of 0.005 s. Table 5 gives the resulting endpoint and time.

Table 4: Prescribed initial conditions.

| Kick | $v_0$ (m/s) | $\beta$ (°) | $\boldsymbol{\omega}_0$ (rad/s) |
|---|---|---|---|
| Power | 27 | 8 | $(0, 0, 0)$ |
| Placement | 20 | 11 | $(0, 0, 0)$ |
| Curved | 24 | 10 | $(0, 0, 35)$ |
| Topspin | 24 | 12 | $(0, 35, 0)$ |
| Backspin | 24 | 12 | $(0, -35, 0)$ |
| Chip | 14 | 20 | $(0, -15, 0)$ |

Table 5: First goal-plane or ground endpoint for all 24 flights.

| Kick | Model | End | $x$ (m) | $y$ (m) | $z$ (m) | $t$ (s) |
|---|---|---|---|---|---|---|
| Power | M1 | goal | 11.000 | 0.000 | 0.826 | 0.4114 |
| Power | M2 | goal | 11.000 | 0.000 | 0.741 | 0.4420 |
| Power | M3 | goal | 11.000 | 0.000 | 0.741 | 0.4420 |
| Power | M4 | goal | 11.000 | 0.000 | 0.776 | 0.4296 |
| Placement | M1 | goal | 11.000 | 0.000 | 0.708 | 0.5603 |
| Placement | M2 | goal | 11.000 | 0.000 | 0.551 | 0.6020 |
| Placement | M3 | goal | 11.000 | 0.000 | 0.551 | 0.6020 |
| Placement | M4 | goal | 11.000 | 0.000 | 0.617 | 0.5848 |
| Curved | M1 | goal | 11.000 | 0.000 | 0.987 | 0.4654 |
| Curved | M2 | goal | 11.000 | 0.000 | 0.879 | 0.5000 |
| Curved | M3 | goal | 11.000 | 0.943 | 0.876 | 0.5027 |
| Curved | M4 | goal | 11.000 | 0.892 | 0.882 | 0.5008 |
| Topspin | M1 | goal | 11.000 | 0.000 | 1.371 | 0.4686 |
| Topspin | M2 | goal | 11.000 | 0.000 | 1.261 | 0.5036 |
| Topspin | M3 | goal | 11.000 | 0.000 | 0.326 | 0.4998 |
| Topspin | M4 | goal | 11.000 | 0.000 | 0.382 | 0.4983 |
| Backspin | M1 | goal | 11.000 | 0.000 | 1.371 | 0.4686 |
| Backspin | M2 | goal | 11.000 | 0.000 | 1.261 | 0.5036 |
| Backspin | M3 | goal | 11.000 | 0.000 | 2.217 | 0.5130 |
| Backspin | M4 | goal | 11.000 | 0.000 | 2.173 | 0.5108 |
| Chip | M1 | goal | 11.000 | 0.000 | 0.684 | 0.8361 |
| Chip | M2 | goal | 11.000 | 0.000 | 0.328 | 0.8994 |
| Chip | M3 | goal | 11.000 | 0.000 | 1.221 | 0.9156 |
| Chip | M4 | goal | 11.000 | 0.000 | 0.368 | 0.9489 |

### 4.1.2 Case 1: Power shot

A high launch speed leaves little time for downward gravitational deflection. Drag removes forward momentum and increases the time required to reach the goal. In the zero-spin condition, M2 and M3 coincide, while M4 changes drag with the decelerating airflow.

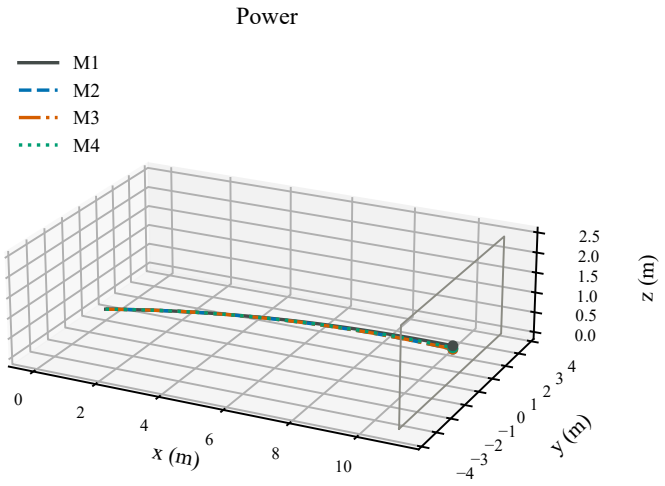



Figure 7: Power shot under the same prescribed initial condition in M1–M4. Endpoint markers denote goal crossing or ground contact.

### 4.1.3 Case 2: Placement shot

The lower launch speed gives gravity more time to bend the path. The comparison shows how the assumed drag coefficient changes both the travel time and crossing height. The initial direction is common to all models.

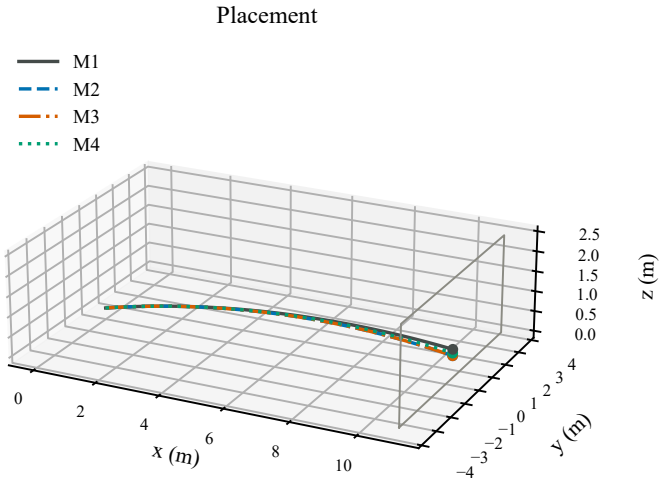



Figure 8: Placement shot with zero spin. M2 and M3 have identical forces and overlapping trajectories.

### 4.1.4 Case 3: Curved shot

Vertical-axis rotation creates a horizontal Magnus force. M1 and M2 preserve the initial horizontal direction; M3 and M4 develop lateral curvature. Their different coefficient descriptions change the amount of bending. The M4 goal crossing is $(0.8921, 0.8822)$ m at $0.5008$ s. This case also supplies the reference state for the rotational-damping comparison in Section 6.2.

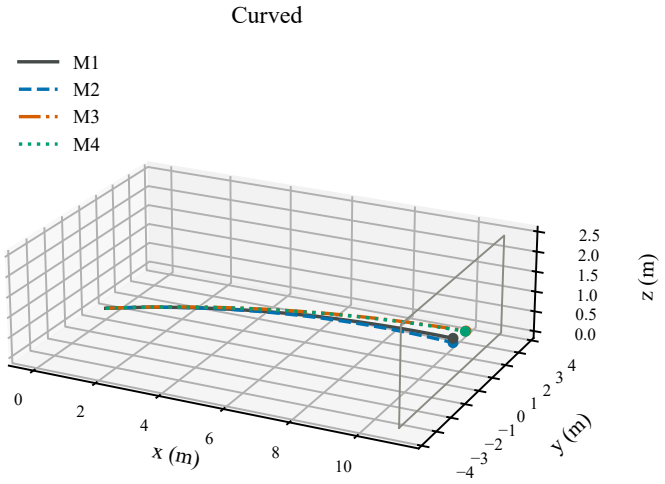



Figure 9: Sidespin redirects the ball horizontally. The four models share speed, elevation and initial spin.

### 4.1.5 Case 4: Topspin and backspin

Changing the sign of $\omega_y$ reverses the vertical Magnus force. Topspin increases downward curvature; backspin opposes the gravitational fall and raises the trajectory. Equal launch speed and angle make this contrast visible directly. Table 5 records each model's actual termination.

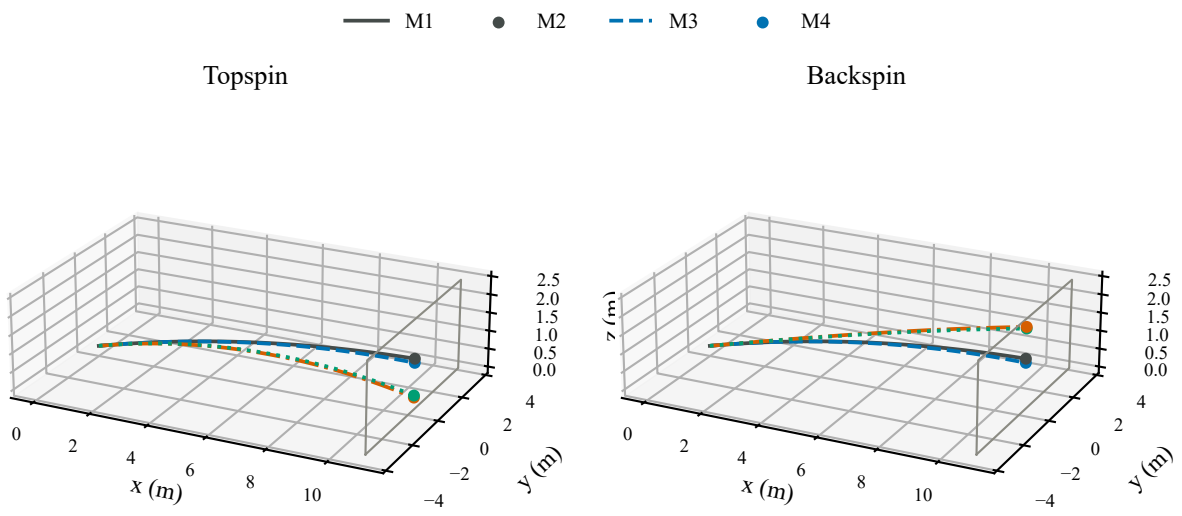



Figure 10: Matched topspin and backspin examples. Only the sign of transverse spin changes between the two panels.

### 4.1.6 Case 5: Chip shot

The chip combines a larger elevation angle, lower speed and backspin. The initial vertical velocity builds an arc, while gravity, drag and upward Magnus lift determine how much height remains near the goal. Its slower airflow also makes the speed-dependent drag transition more relevant.

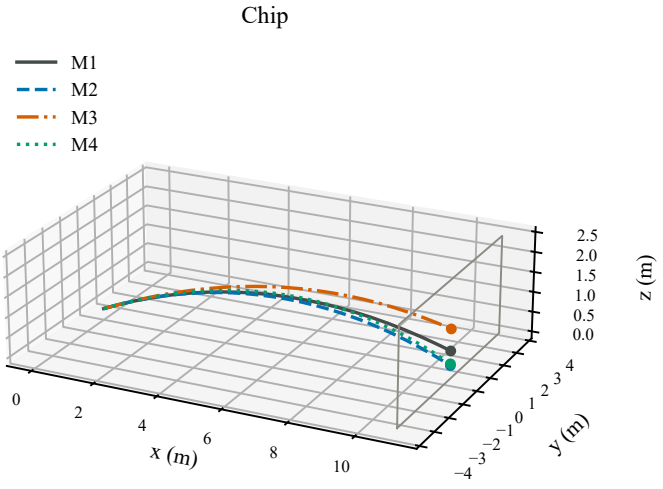



Figure 11: The prescribed chip illustrates the combined effect of launch angle, slower airflow and backspin.

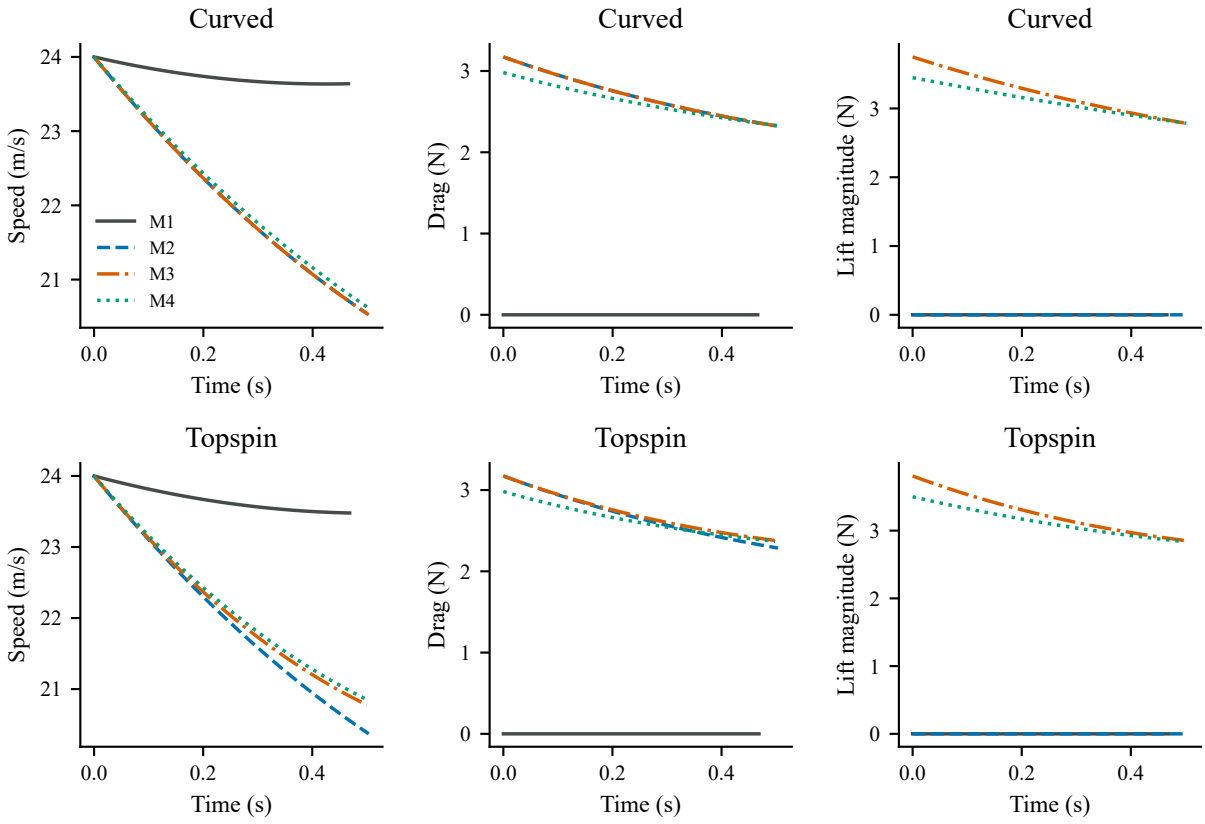



Figure 12: Speed, drag and lift magnitudes for the curved and topspin examples. Each curve ends at that model's first goal crossing or ground contact.

The force histories connect the paths to their causes. Dynamic pressure decreases as drag removes speed. A constant coefficient still gives a changing force through the $v^2$ factor. In M4, changing $Re$ and $Sp$ additionally changes the coefficients. The spin direction decides whether lift acts chiefly sideways or vertically.

## 4.2 Inverse Prediction from Three Real Penalties

### 4.2.1 Measurements, coordinate system and input times

The Sportec Open DFL data associate actual match events with TRACAB Gen5 multi-camera tracking at 25 Hz [6, 7]. We use penalties by Dejan Ljubicic (J03WMX), Dawid Igor Kownacki (J03WOH), and Philipp Klement (J03WR9). The supplied CSV retains 39 source frames, respectively 15, 12 and 12. Positions are used as published; no added observation-noise model is introduced.

The source horizontal origin is the pitch centre. All three shots travel towards $X = 52.5$ m. To use the competition's penalty-mark coordinates, we apply

$$x = X - 41.5, \qquad y = Y, \qquad z = Z_{\text{original}}, \qquad x_g = 11 \text{ m}. \tag{19}$$

The nominal penalty mark is the horizontal origin, so each actual first observation retains its measured offset. Released $Z$ values are interpreted as metres and ball-centre height. Mass, radius and air properties use the model reference settings in Section 2; the supplied source leaves per-ball mass and radius unreported.

Each inversion defines $t = 0$ at its selected source frame. J03WMX uses samples 3, 4 and 5 of the original sequence; J03WOH and J03WR9 use samples 2, 3 and 4. All three fitting times are 0, 0.04 and 0.08 s. Their time origins on the original recording are 0.08, 0.04 and 0.04 s, respectively. Table 6 gives the measured starting positions. Earlier records, repeated frames and post-goal records remain in the raw CSV. Trajectory comparison begins at the selected start, with subsequent samples used only for comparison up to the observed goal crossing. Frame-interpolated crossings are marked separately from measured frames.

Table 6: Frame counts, fitting times and measured local starting positions.

| Event | Frames | Fit times (s) | $\boldsymbol{r}_0$ (m) |
|---|---|---|---|
| J03WMX | 15 | 0.00,0.04,0.08 | $(3.04, 0.55, 0.13)$ |
| J03WOH | 12 | 0.00,0.04,0.08 | $(0.77, 0.08, 0.24)$ |
| J03WR9 | 12 | 0.00,0.04,0.08 | $(1.47, 0.49, 0.23)$ |

### 4.2.2 Bounded inversion and consistent selection

For observed positions $\boldsymbol{p}_1, \boldsymbol{p}_2$ at $t_1, t_2$, the position objective is

$$J_r = \sum_{i=1}^{2} \|\boldsymbol{r}(t_i; \boldsymbol{\eta}) - \boldsymbol{p}_i\|^2, \qquad \boldsymbol{\eta}_{\text{M4}} = (\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L). \tag{20}$$

Next we favor force corrections near the published scale,

$$J_\alpha = (\alpha_D - 1)^2 + (\alpha_L - 1)^2, \tag{21}$$

using the multipliers present in each model. Among equivalent M4 states, the smallest spin magnitude selects a representative. The numerical objective allowances are $6 \times 10^{-12}$ m$^2$ for $J_r$ and $10^{-8}$ for $J_\alpha$.

The implemented search uses the component constraints in Table 7. These are declared analysis settings. The speed and spin scales refer to the empirical and experimental ranges [12, 11]; the component boxes and the force-correction interval specify this inverse calculation. M3's unit spin axis carries direction only.

Table 7: Component bounds and force-correction settings for inversion.

| Quantity | Search interval |
|---|---|
| $v_{x0}$ | $[0, 35]$ m/s |
| $v_{y0}, v_{z0}$ | $[-35, 35]$ m/s |
| $\omega_{x0}, \omega_{y0}, \omega_{z0}$ (M4) | $[-20\pi, 20\pi]$ rad/s |
| Azimuth, elevation (M3) | $[-\pi, \pi], [-\pi/2, \pi/2]$ rad |
| $\alpha_D, \alpha_L$ | $[0.5, 1.5]$ |

M2 uses one displacement-based start, M3 uses six axis-direction starts, and M4 uses 19 starts including zero spin and signed coordinate-axis spins. Least squares allows 240 function evaluations per start. Coefficient and spin selection use the same candidate state when evaluating the residual and the secondary objective. The fitted state and configuration then enter the same dynamics for prediction.

### 4.2.3 Predicted trajectories, initial states and goal crossings

For each penalty we list the three measured fitting positions and observed goal crossing, followed by the initial velocity, spin information and predicted crossing from each model. M1 and M2 do not use spin; M3 reports its unit spin axis and M4 reports the three angular-velocity components. The deviation of a predicted crossing from the observed crossing is

$$d_g = \sqrt{(y_{\text{pred}} - y_{\text{obs}})^2 + (z_{\text{pred}} - z_{\text{obs}})^2} \quad \text{m}. \tag{22}$$

Deviation is evaluated at each model's own goal-plane crossing. Models terminating on the ground retain their termination status, with crossing and deviation shown as "—. Open circles mark fit inputs, squares mark ground contact, and an asterisk identifies an iteration-limit candidate. All model paths stop at first ground contact.

**J03WMX**

Table 8: J03WMX: three measured fitting points and observed goal crossing (direct source frame).

| Point | Source frame | $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|---|---|
| $P_0$ | 152602 | 0.00 | 3.040 | 0.550 | 0.130 |
| $P_1$ | 152603 | 0.04 | 4.190 | 0.820 | 0.150 |
| $P_2$ | 152604 | 0.08 | 5.250 | 1.070 | 0.180 |
| Goal | 152610 | 0.3200 | 11.000 | 2.7500 | 0.1600 |

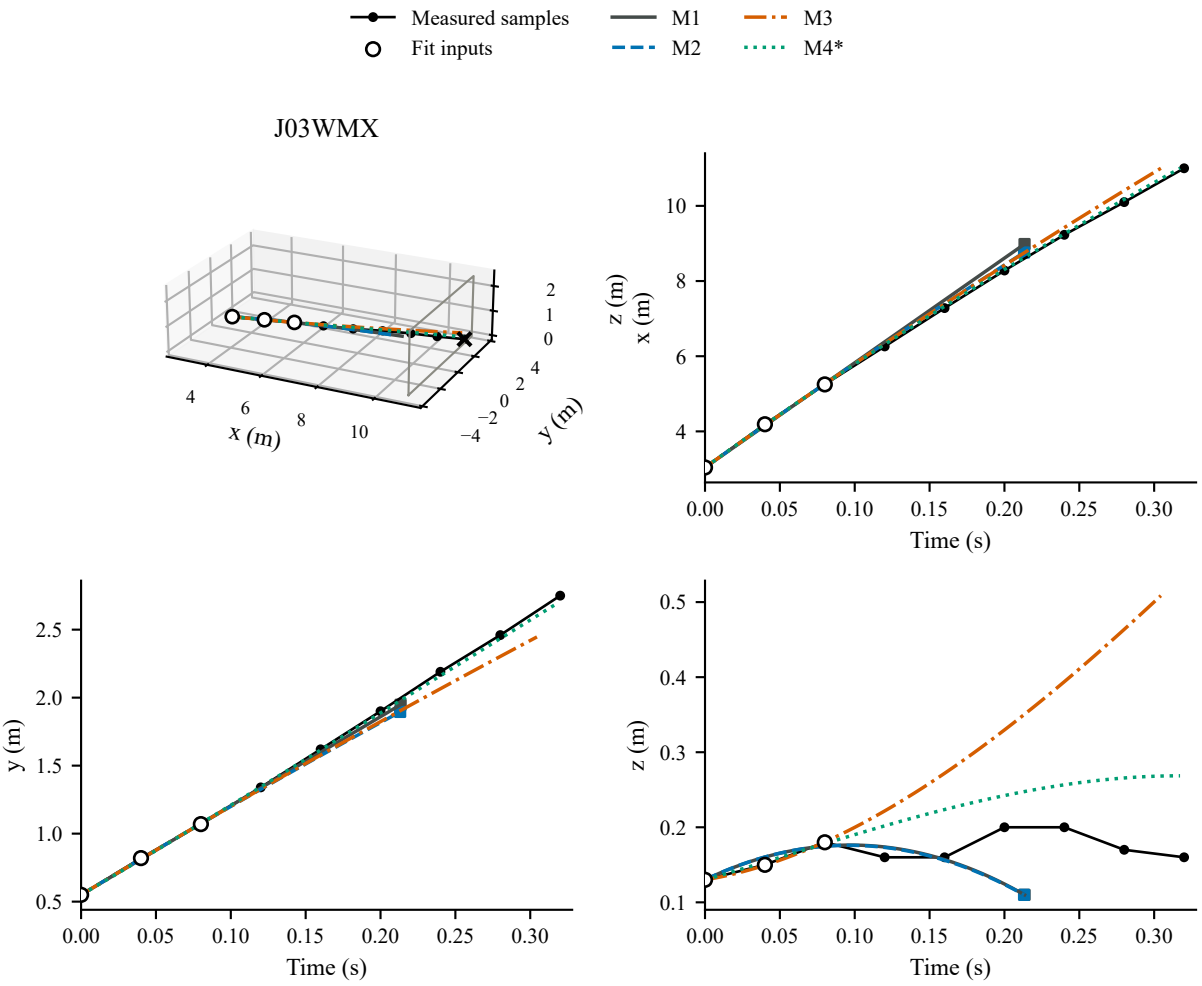



Figure 13: J03WMX: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 9: J03WMX: inverse initial states and predicted goal-plane crossings. Spin entries have units rad/s in M4; the starred M3 vector is a unit axis. M1/M2 do not estimate spin. Limit rows retain the iteration-limit candidate.

| Model | Status | $y_g$ (m) | $z_g$ (m) | $\boldsymbol{v}_0$ (m/s) | Spin | $d_g$ (m) |
|---|---|---|---|---|---|---|
| M1 | ground | — | — | $\begin{pmatrix} 27.850 \\ 6.550 \\ 0.953 \end{pmatrix}$ | — | — |
| M2 | ground | — | — | $\begin{pmatrix} 28.411 \\ 6.682 \\ 0.967 \end{pmatrix}$ | — | — |
| M3 | plane | 2.4464 | 0.5086 | $\begin{pmatrix} 28.428 \\ 6.659 \\ 0.327 \end{pmatrix}$ | $\begin{pmatrix} 0.092 \\ -0.995 \\ 0.038 \end{pmatrix}^{*}$ | 0.4622 |
| M4 | limit | 2.6884 | 0.2686 | $\begin{pmatrix} 28.776 \\ 6.431 \\ 0.574 \end{pmatrix}$ | $\begin{pmatrix} 62.832 \\ -62.832 \\ 62.832 \end{pmatrix}$ | 0.1248 |

The measured low flight requires sustained vertical support from lift. M4's crossing deviation is 0.1248 m. The iteration-limit candidates are M4. Ground contact precedes the goal for M1, M2.

**J03WOH**

Table 10: J03WOH: three measured fitting points and observed goal crossing (linear interpolation).

| Point | Source frame | $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|---|---|
| $P_0$ | 128955 | 0.00 | 0.770 | 0.080 | 0.240 |
| $P_1$ | 128956 | 0.04 | 1.990 | 0.340 | 0.460 |
| $P_2$ | 128957 | 0.08 | 3.180 | 0.620 | 0.650 |
| Goal | 128964–128965 | 0.3643 | 11.000 | 2.9188 | 1.1789 |

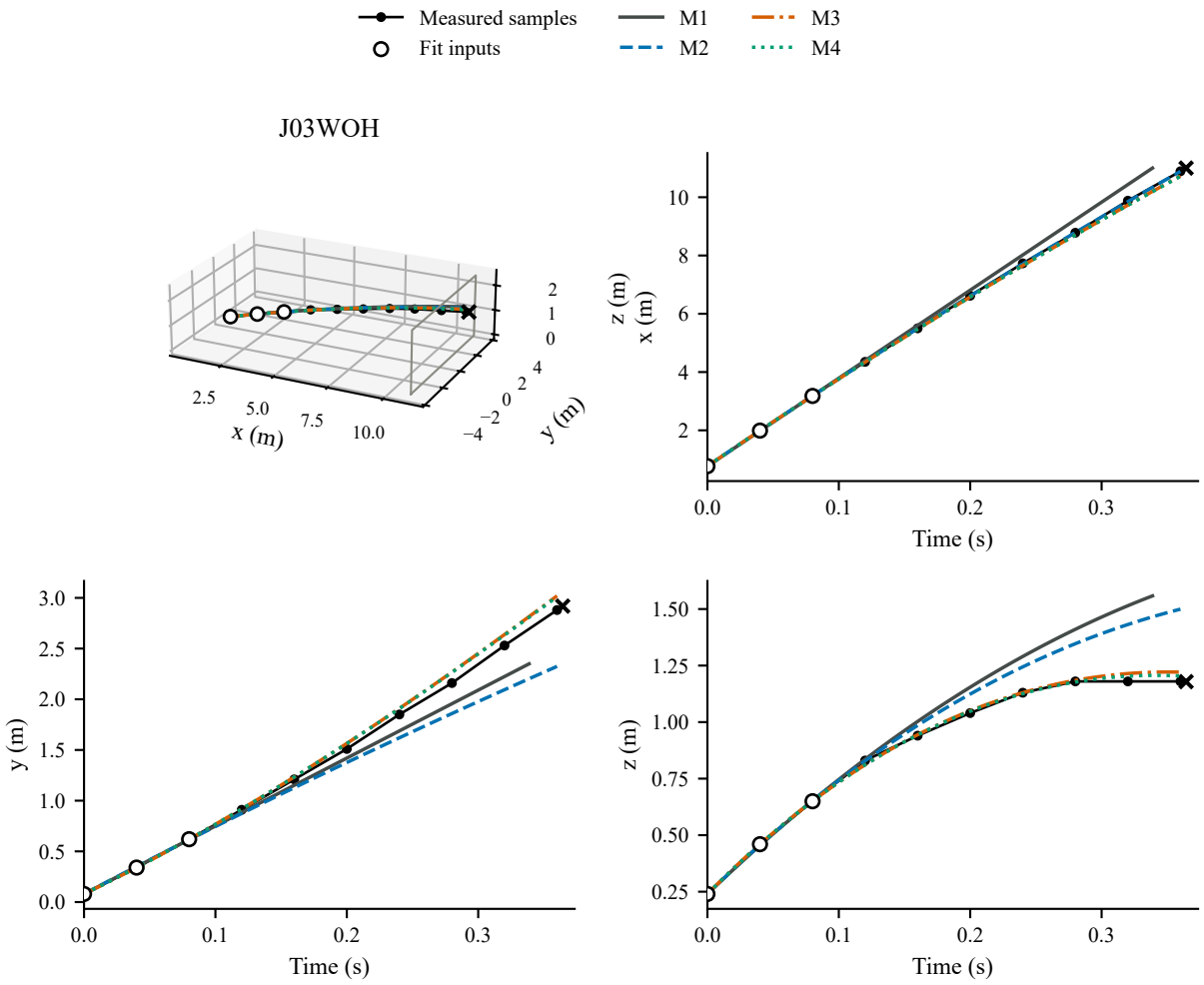



Figure 14: J03WOH: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 11: J03WOH: inverse initial states and predicted goal-plane crossings. Spin entries have units rad/s in M4; the starred M3 vector is a unit axis. M1/M2 do not estimate spin. Limit rows retain the iteration-limit candidate.

| Model | Status | $y_g$ (m) | $z_g$ (m) | $\boldsymbol{v}_0$ (m/s) | Spin | $d_g$ (m) |
|---|---|---|---|---|---|---|
| M1 | plane | 2.3496 | 1.5583 | $\begin{pmatrix}30.200\\6.700\\5.553\end{pmatrix}$ | — | 0.6841 |
| M2 | plane | 2.3494 | 1.5059 | $\begin{pmatrix}30.794\\6.831\\5.658\end{pmatrix}$ | — | 0.6567 |
| M3 | plane | 3.1269 | 1.2184 | $\begin{pmatrix}30.881\\6.236\\5.884\end{pmatrix}$ | $\begin{pmatrix}-0.228\\0.317\\0.921\end{pmatrix}^{*}$ | 0.2118 |
| M4 | plane | 3.1221 | 1.1997 | $\begin{pmatrix}30.879\\6.235\\5.882\end{pmatrix}$ | $\begin{pmatrix}-39.084\\17.890\\62.832\end{pmatrix}$ | 0.2044 |

The rising measured arc connects vertical launch velocity with the subsequent action of gravity and lift. M4's crossing deviation is 0.2044 m.

**J03WR9**

Table 12: J03WR9: three measured fitting points and observed goal crossing (linear interpolation).

| Point | Source frame | $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|---|---|
| $P_0$ | 176244 | 0.00 | 1.470 | 0.490 | 0.230 |
| $P_1$ | 176245 | 0.04 | 2.570 | 0.850 | 0.340 |
| $P_2$ | 176246 | 0.08 | 3.640 | 1.200 | 0.410 |
| Goal | 176253–176254 | 0.3695 | 11.000 | 3.0697 | 0.4657 |

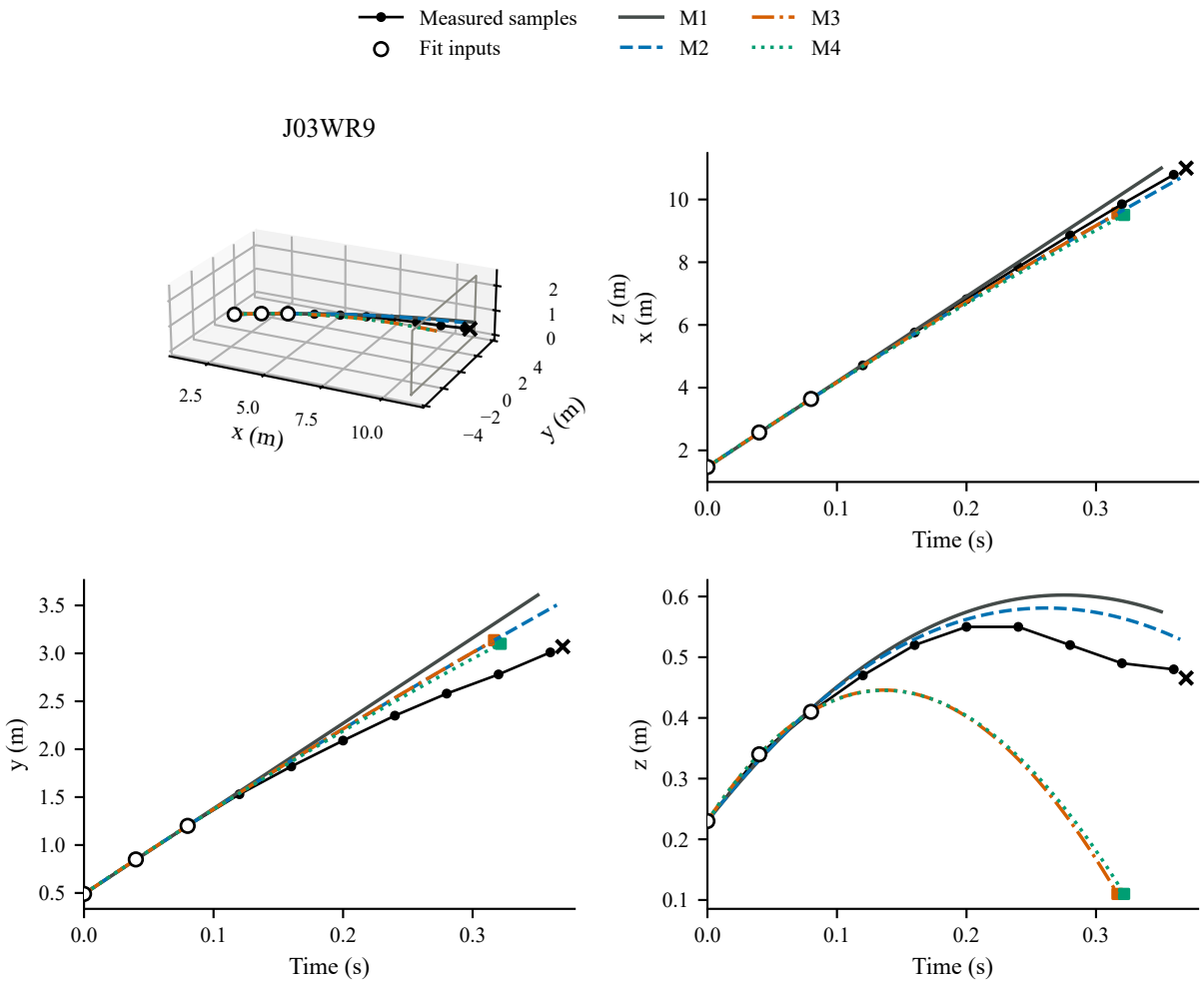



Figure 15: J03WR9: predicted and sampled trajectories from the selected start; fitting inputs are at 0, 0.04 and 0.08 s.

Table 13: J03WR9: inverse initial states and predicted goal-plane crossings. Spin entries have units rad/s in M4; the starred M3 vector is a unit axis. M1/M2 do not estimate spin. Limit rows retain the iteration-limit candidate.

| Model | Status | $y_g$ (m) | $z_g$ (m) | $\boldsymbol{v}_0$ (m/s) | Spin | $d_g$ (m) |
|---|---|---|---|---|---|---|
| M1 | plane | 3.6083 | 0.5750 | $\begin{pmatrix}27.200\\8.900\\2.703\end{pmatrix}$ | — | 0.5496 |
| M2 | plane | 3.6083 | 0.5147 | $\begin{pmatrix}27.747\\9.079\\2.753\end{pmatrix}$ | — | 0.5408 |
| M3 | ground | — | — | $\begin{pmatrix}27.703\\9.070\\3.252\end{pmatrix}$ | $\begin{pmatrix}-0.174\\0.985\\0.003\end{pmatrix}^*$ | — |
| M4 | ground | — | — | $\begin{pmatrix}27.881\\9.124\\3.271\end{pmatrix}$ | $\begin{pmatrix}-62.832\\62.832\\-3.805\end{pmatrix}$ | — |

The measured path bends laterally, so fitted spin direction affects the continuation towards the goal. M4 reaches the ground at $x = 9.507$ m. Ground contact precedes the goal for M3, M4.

# 5 Sensitivity Analysis

Section 4 illustrates flight prediction through prescribed states and actual penalty records. We now examine how parameter changes affect the goal-plane crossing. The competition's three positions provide a separate reference example, with mass fixed at $m = 0.43$ kg, radius at $R = 0.11$ m, and the same constant-spin, still-air M4 coefficient relations.

Table 14: Early positions for the sensitivity example.

| $t$ (s) | $x$ (m) | $y$ (m) | $z$ (m) |
|---|---|---|---|
| 0 | 0 | 0 | 0.11 |
| 0.05 | 1.50 | 0.45 | 0.44 |
| 0.10 | 2.96 | 0.86 | 0.74 |

The reference state is

$$\begin{aligned}\boldsymbol{v}_0 &= (30.4063, 9.4256, 6.9064)\ \mathrm{m/s},\\ \boldsymbol{\omega}_0 &= (-49.0903, -31.4834, -112.2656)\ \mathrm{rad/s}.\end{aligned} \tag{23}$$

with $\alpha_D = 1.0001$ and $\alpha_L = 1.0000$. Calculations use the saved full-precision state, compatible with Table 14. Its crossing is $(y_g^*, z_g^*) = (2.6826, 1.9592)$ m at $t_g^* = 0.402649$ s. We examine individual perturbations, states refitted to the same early positions, and joint launch perturbations.

## 5.1 Effects of individual parameters

One parameter is changed at a time while the other conditions retain their reference values. Launch speed varies by $\pm 5\%$; spin magnitude, drag and lift multipliers, and air density each vary by $\pm 10\%$;

velocity and spin-axis azimuth and elevation each vary by $\pm 2^\circ$. Each parameter has 41 equally spaced values, giving 369 trajectories. Speeds remain below the empirical relation's 35 m/s upper limit, and all trajectories reach the goal plane before ground contact.

For the reference crossing $(y_g^*, z_g^*, t_g^*)$, define

$$\Delta y_g = y_g - y_g^*, \quad \Delta z_g = z_g - z_g^*, \quad \Delta t_g = t_g - t_g^*, \qquad D_g = \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \tag{24}$$

Table 15 compares scalar changes of $\pm 5\%$ and direction changes of $\pm 1^\circ$, reporting the larger crossing displacement and arrival-time change from each positive–negative pair.

Table 15: Goal-plane displacement and arrival-time change under one-factor perturbations.

| Parameter | Perturbation | max $D_g$ (cm) | max $\lvert\Delta t_g\rvert$ (ms) |
|---|---|---|---|
| Launch speed | $\pm 5\%$ | 10.796 | 23.187 |
| Velocity azimuth | $\pm 1^\circ$ | 20.179 | 1.943 |
| Velocity elevation | $\pm 1^\circ$ | 20.247 | 1.922 |
| Spin magnitude | $\pm 5\%$ | 6.239 | 2.072 |
| Spin-axis azimuth | $\pm 1^\circ$ | 0.675 | 0.049 |
| Spin-axis elevation | $\pm 1^\circ$ | 1.049 | 0.115 |
| Drag multiplier | $\pm 5\%$ | 0.859 | 2.664 |
| Lift multiplier | $\pm 5\%$ | 3.584 | 0.263 |
| Air density | $\pm 5\%$ | 3.122 | 2.404 |

Figure 16 gives the lateral and vertical responses. To compare arrival times on common axes, Figure 17 divides each perturbation by its scan half-width: $\pm 1$ corresponds to speed changes of $\pm 5\%$, other scalar changes of $\pm 10\%$, or angle changes of $\pm 2^\circ$. The vertical axis retains milliseconds.

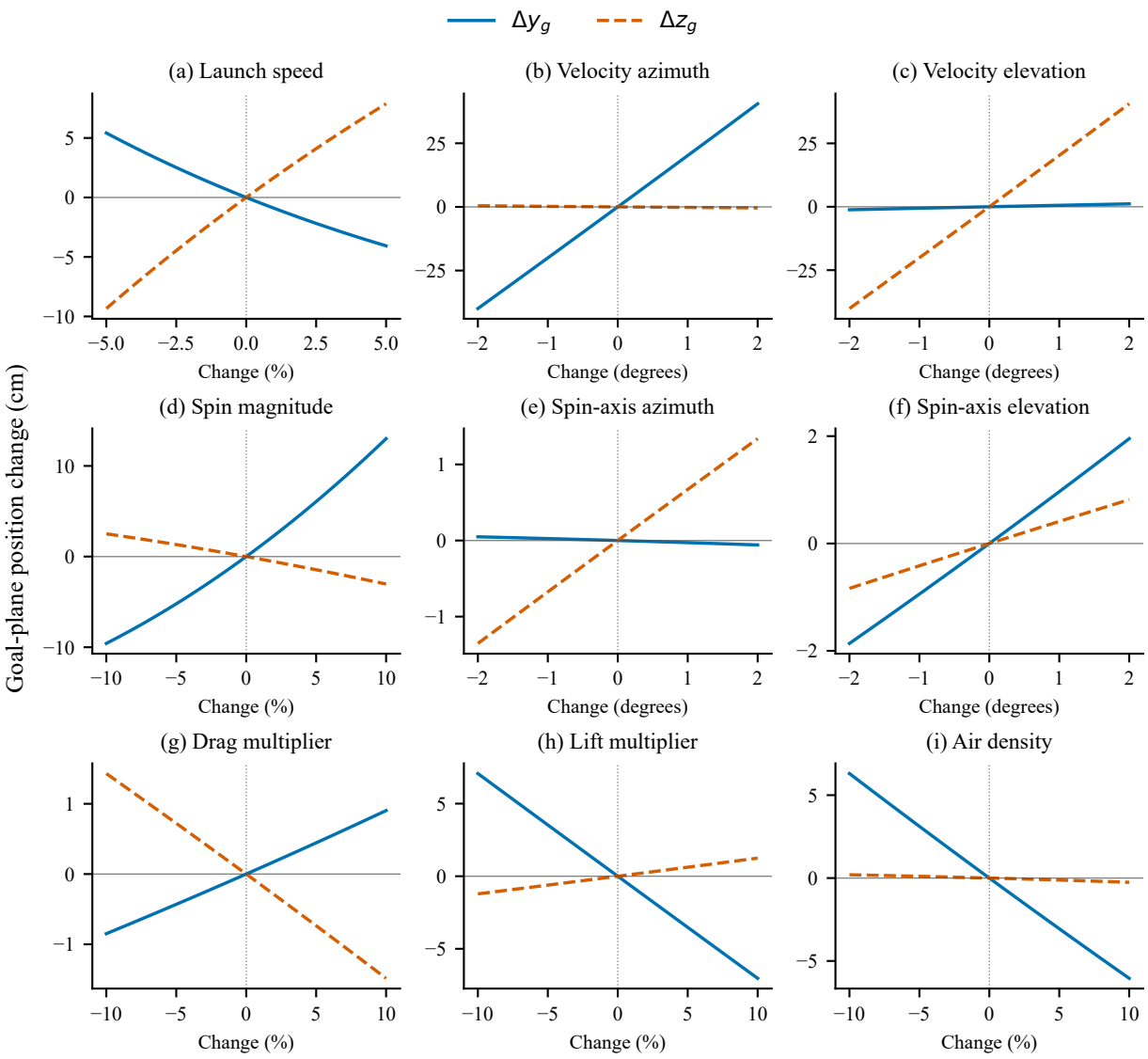



Figure 16: Goal-plane position responses to nine one-factor perturbations. Blue solid curves show $\Delta y_g$ and orange dashed curves show $\Delta z_g$, in centimetres. Scalar changes are percentages; direction changes are in degrees. Ball mass and radius remain fixed. Vertical scales are set separately for each panel.

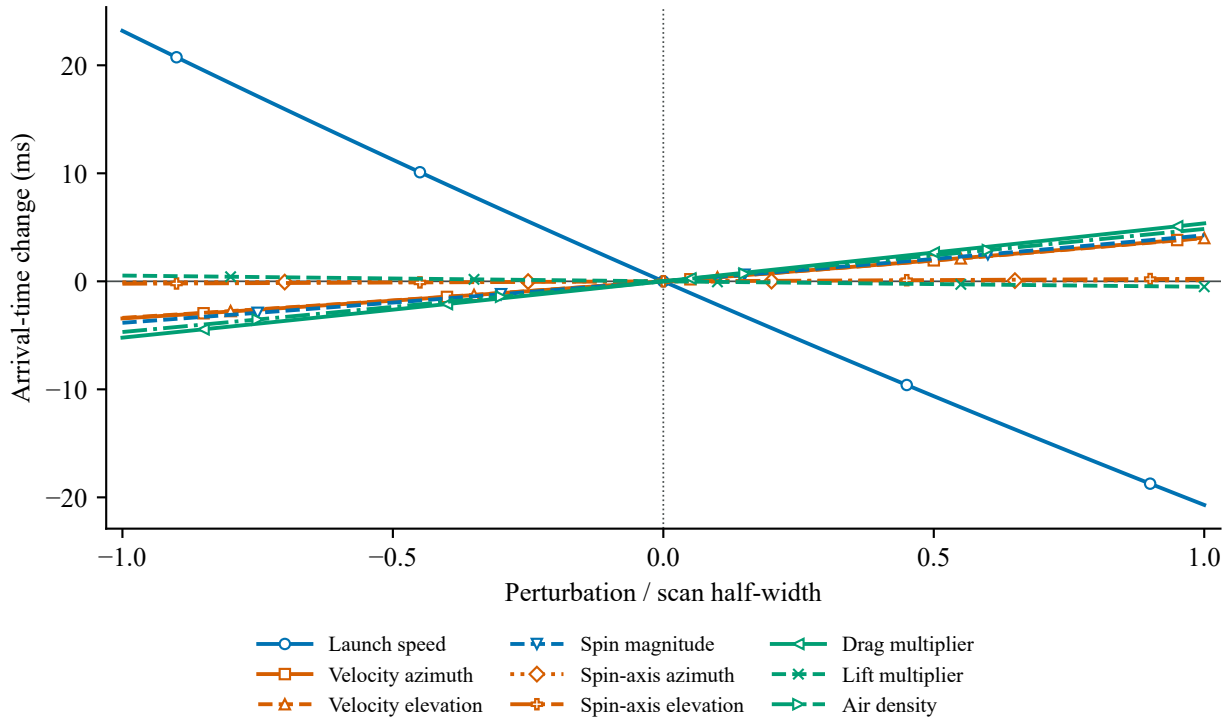



Figure 17: One-factor arrival-time responses. Perturbations are normalized by scan half-width and $\Delta t_g$ is in milliseconds. Positive values indicate later arrival and negative values earlier arrival; each parameter has 41 values.

A $1^\circ$ change in velocity azimuth or elevation moves the crossing by about 0.20 m, mainly in the corresponding lateral or vertical direction. A 5% speed change gives a maximum displacement of 0.108 m and an arrival-time change of 23.187 ms. Higher speed shortens the action of gravity and aerodynamic forces. Lift scaling primarily changes transverse curvature, while drag scaling affects arrival time through deceleration.

Spin has an additional nonlinear effect through the complete coefficient relation. In its positive-lift region, at fixed speed,

$$C_L \propto Sp^{0.83}\left[C_{D,\text{ceil}} - f(v) - 0.514Sp\right]. \tag{25}$$

At the reference launch speed, this expression peaks at $Sp \simeq 0.3007$, below the reference initial value 0.4272. Increasing spin therefore reduces the empirical lift coefficient near this state. Since the reference Magnus force bends the ball toward smaller $y$, a 5% spin increase gives $y_g = 2.7433$ m, whereas a 5% decrease gives $y_g = 2.6305$ m. Spin and lift scaling can consequently compensate for one another.

## 5.2 Crossing variation under early-position constraints

The same early positions can arise from different combinations of initial velocity, spin and aerodynamic forces. Two later positions constrain six coordinates, while M4 contains eight inverse quantities. After selecting a representative state, we therefore examine how other compatible states change the goal-plane crossing.

We take a $17 \times 17$ grid of $(\alpha_D, \alpha_L)$ in $[0.8, 1.2]^2$ and refit the initial velocity and spin at each point while retaining the positions at 0.05 and 0.10 s. The search uses total speed at most 35 m/s, total spin at most 200 rad/s, and a maximum coordinate residual of $10^{-7}$ m. The spin range includes the representative state and defines the conditions for this comparison.

Grid searches and local constrained optimization give Table 16. The nested interval $[0.9, 1.1]^2$ shows how the crossing ranges change as the force-multiplier range expands.

Table 16: Ranges of goal-plane crossings compatible with the early positions.

| Range of each multiplier | $y_g$ (m) | $z_g$ (m) | $t_g$ (s) |
|---|---|---|---|
| $[0.9, 1.1]$ | 2.6448–2.7719 | 1.9306–1.9734 | 0.4023–0.4034 |
| $[0.8, 1.2]$ | 2.6237–2.7980 | 1.9266–1.9827 | 0.4021–0.4036 |

For multipliers in $[0.8, 1.2]^2$, the largest crossing displacement found relative to the representative state is 0.1183 m, and all compatible trajectories clear the goal. Figure 18 shows the lateral–vertical pairs, quantifying the remaining variation under the same early-position constraints.

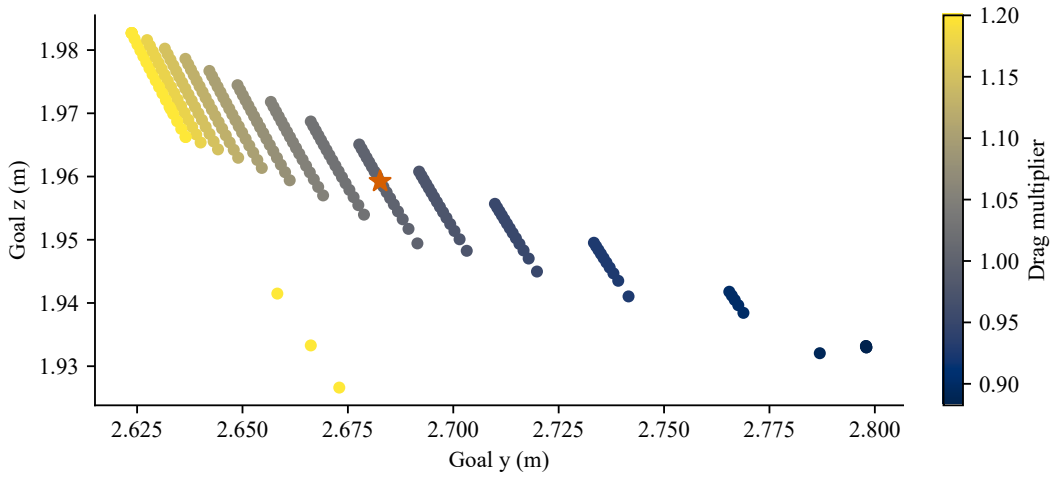



Figure 18: Crossings compatible with the same early positions, using the wider force-multiplier setting. Colour denotes the drag multiplier; the orange star is the reference state.

Air density provides a further example of parameter compensation. With $m, R, \nu$ fixed, changing density from $\rho$ to $\rho'$ and taking

$$\alpha_D' = \alpha_D \frac{\rho}{\rho'}, \qquad \alpha_L' = \alpha_L \frac{\rho}{\rho'} \tag{26}$$

leaves both aerodynamic accelerations and the entire trajectory unchanged. Density changes with fixed force multipliers therefore give a different response from changes accompanied by a refit: the early positions constrain the combined aerodynamic action of these parameters.

## 5.3 Joint perturbations and Monte Carlo simulation

Several flight conditions can change together. To examine their joint effect, we independently sample uniform perturbations over the full ranges above: speed $\pm 5\%$, each direction angle $\pm 2^\circ$, and spin magnitude, both force multipliers and density $\pm 10\%$, with $\rho \in [1.08, 1.32]$ kg/m$^3$. Seed 20261108 generates 100000 trajectories. Mass, radius and initial position remain fixed, while the early positions vary with the launch state; Section 5.2 holds the observed positions fixed.

All simulated trajectories reach the goal plane before ground contact. Table 17 reports the sampled ranges and the 2.5th–97.5th percentiles of each position coordinate and arrival time. The mean crossing is $(2.6927, 1.9561)$ m, with lateral and vertical standard deviations 0.2494 m and 0.2385 m. The largest displacement from the representative crossing is 0.835 m.

Table 17: Monte Carlo crossing positions and arrival times (100,000 trajectories).

| Output | Sample minimum–maximum | Central 95% interval, one coordinate |
|---|---|---|
| $y_g$ (m) | 2.0308–3.3515 | 2.2508–3.1364 |
| $z_g$ (m) | 1.4055–2.4868 | 1.5494–2.3664 |
| $t_g$ (s) | 0.3687–0.4485 | 0.3795–0.4293 |

For a joint spatial region, let $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^T$, with sample mean $\bar{\boldsymbol{q}}$ and covariance $\widehat{\Sigma}$. Compute

$$\begin{aligned} d_i^2 &= (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}), \\ \mathcal{E}_{0.95} &= \left\{ \boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95} \right\}, \end{aligned} \tag{27}$$

where $c_{0.95} = 4.6575$ is the empirical 95th percentile of the squared distances. The ellipse in Figure 19 encloses 95% of the simulated crossings, describing lateral and vertical variation jointly, while the intervals in Table 17 describe each coordinate separately.

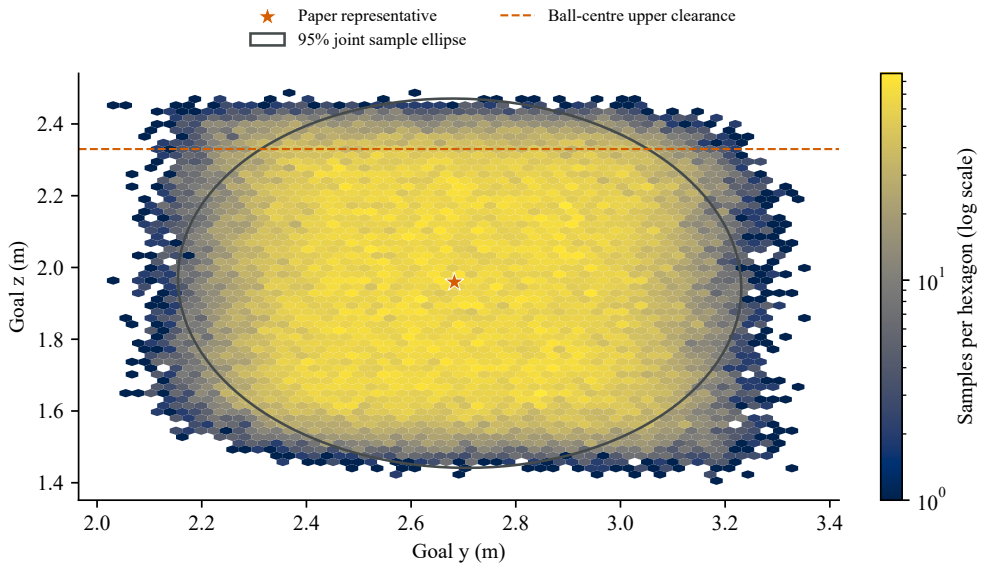



Figure 19: Monte Carlo goal-plane crossings for 100,000 primary samples, with fixed ball mass and radius. Hexagon colour encodes sample count. The ellipse encloses 95% of the samples; the horizontal line is the ball-centre upper clearance $z_g = 2.33$ m.

Under the stated uniform perturbations, the ball-centre clearance criterion $|y_g| \leq 3.66 - R$, $R \leq z_g \leq 2.44 - R$ gives 94870 trajectories (94.87%) passing fully through the opening. The remaining 5130 trajectories (5.13%) are too high.

Figure 20 connects inputs with flight outcomes. A random subset of 400 trajectories is displayed for readability; each polyline passes through nine inputs and three outputs. Here $v_0, \omega_0$ are velocity and spin magnitudes; $\phi, \theta$ are azimuth and elevation, with subscripts $v, \omega$ distinguishing velocity direction and spin axis.

Inputs are scaled by their perturbation limits and outputs by the extrema of all 100,000 trajectories, mapping each axis linearly to 0–1 while retaining its original limits. Blue lines clear the goal, orange lines are too high, and the grey dashed line is the representative state. The 382 blue and 18 orange traces illustrate parameter combinations; the percentages above use the full sample.

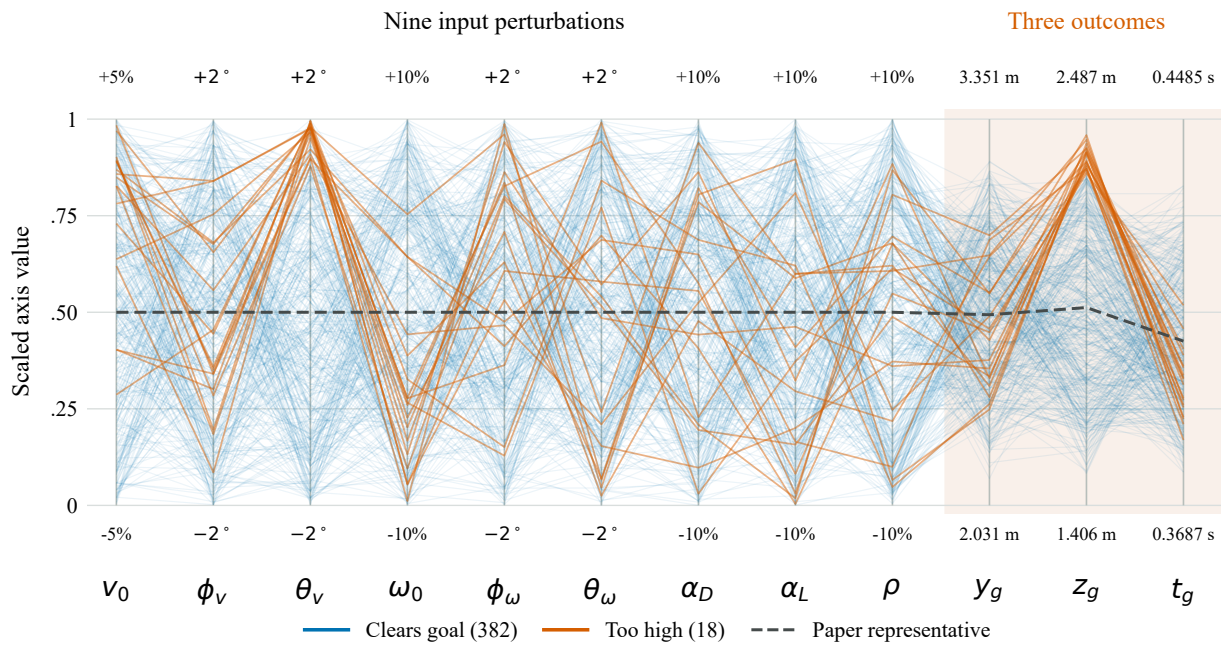



Figure 20: Parallel coordinates linking nine input perturbations to goal-plane positions and arrival time. The first nine axes are inputs; the three shaded axes are outputs. Each coordinate is scaled to 0–1 within its own range. A random subset of 400 trajectories is shown, with colour distinguishing goal-clearance outcomes.

Orange traces favour positive velocity-elevation perturbations on the $\theta_v$ axis and high values on the $z_g$ axis, relating larger launch elevations to overly high crossings in this example. Following a trace also reveals its speed, spin, force corrections and arrival time.

Figure 21 examines increasing sample counts. From 50000 to 100000 trajectories, the central-95% interval endpoints change by at most 0.472 mm laterally, 0.210 mm vertically and 0.155 ms in time. Two independent seeds, each with 10000 trajectories, give similar central intervals.

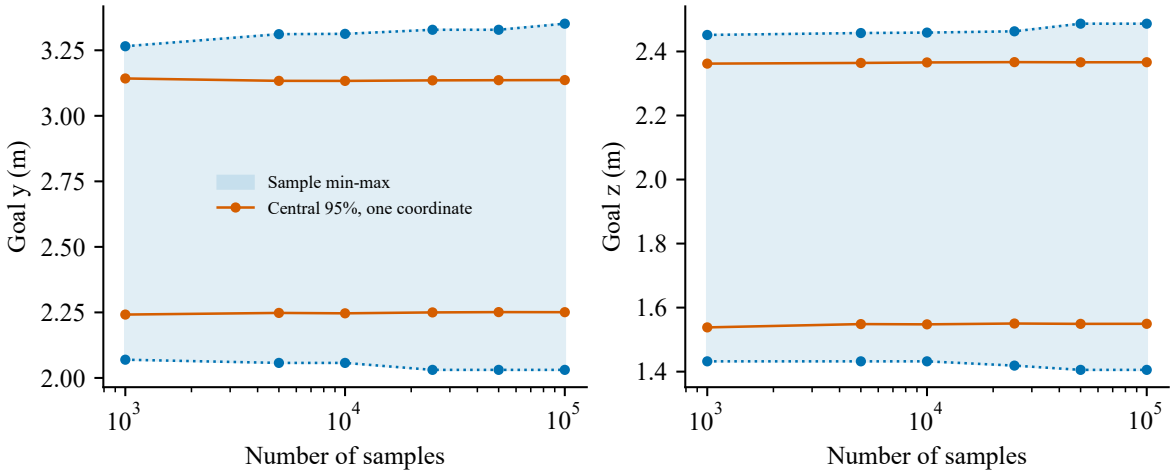



Figure 21: Sample-count checks for the lateral and vertical crossings. Shading and dotted curves show sampled extrema; orange curves show each coordinate's central 95% interval.

The trajectories are integrated with fourth-order Runge–Kutta and checked using independent adaptive integration and a halved step; crossing differences remain below $10^{-6}$ m. Individual responses, compatible trajectories and joint launch perturbations together provide a physical explanation and quantitative ranges for the goal-plane variation.

# 6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

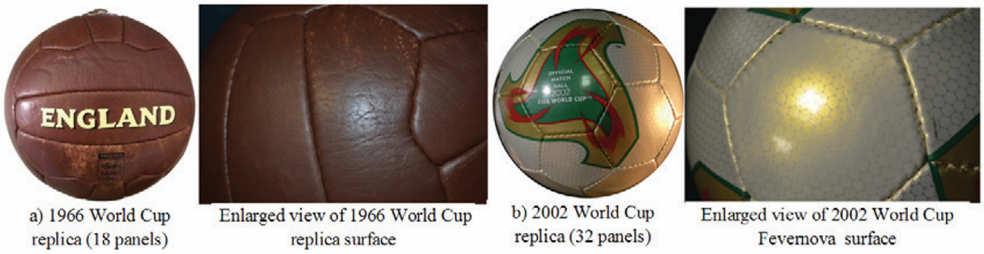
## 6.1 Asymmetric Surface Geometry

Paudel and Yan [9] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



Figure 22: Illustrative comparison of a paneled football and a smooth sphere, inspired by Paudel and Yan [9]. Their compared bodies had diameter 0.22 m and mass 0.436 kg. The panel geometry shown here is schematic.

Alam et al. [16] measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere. Figure 23 shows the panels and surface textures of these footballs, while Figure 24 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.




a) 1966 World Cup replica (18 panels) | Enlarged view of 1966 World Cup replica surface | b) 2002 World Cup replica (32 panels) | Enlarged view of 2002 World Cup Fevernova surface

Fig. 1. Vintage and Adidas Fevernova balls



a) 2006 World Cup replica (14 panels) | Enlarged View of 2006 World Cup Teamgeist II surface | b) 2009 Confederation Cup (14 panels) | Enlarged View of 2009 Confederation Cup Teamgeist III surface

Fig. 2. Adidas Teamgeist 2006 and 2009 balls



a) 2010 World Cup replica (8 panels) | Enlarged view of 2010 World Cup Jabulani surface | b) 2010 Nike T-90 replica (32 panels) | Enlarged view of 2010 Nike T-90 surface

Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

Figure 23: Six footballs and details of their surfaces, reproduced from Figures 1–3 of Alam et al. [16].

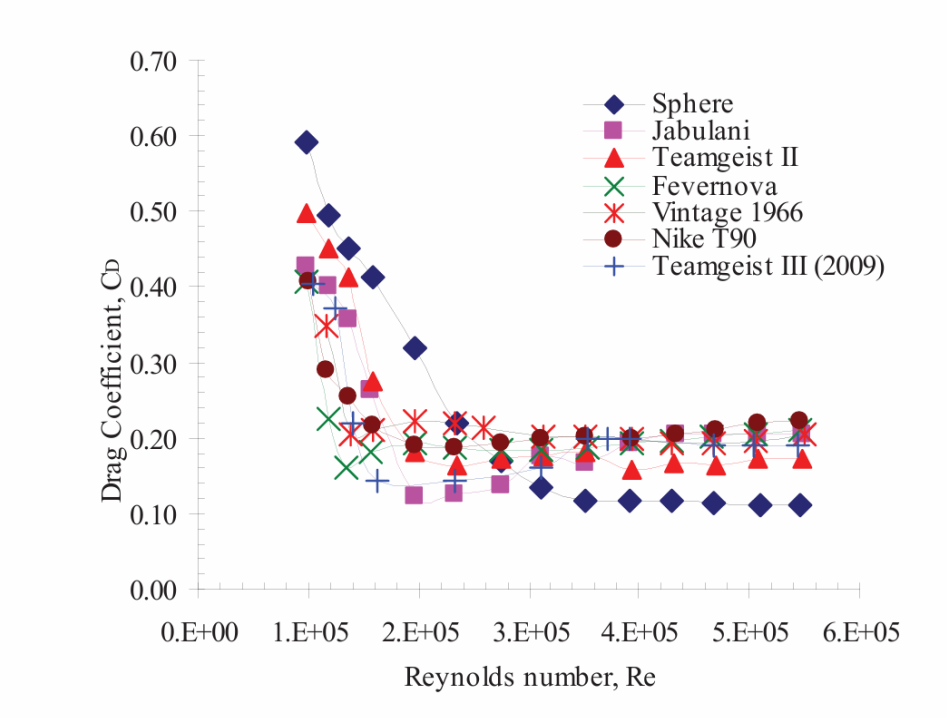



Figure 24: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through $C_D$ and $C_L$. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

## 6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [13], Supplementary Section 8.4, and the MuJoCo ellipsoid model [14], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{28}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [14]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\,Sp}. \tag{29}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{30}$$

It is $3.54 \times 10^{-4}$ at the prescribed curved-shot launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [8], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{31}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

Use the curved shot of Section 4.1, with initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with Eq. (31).

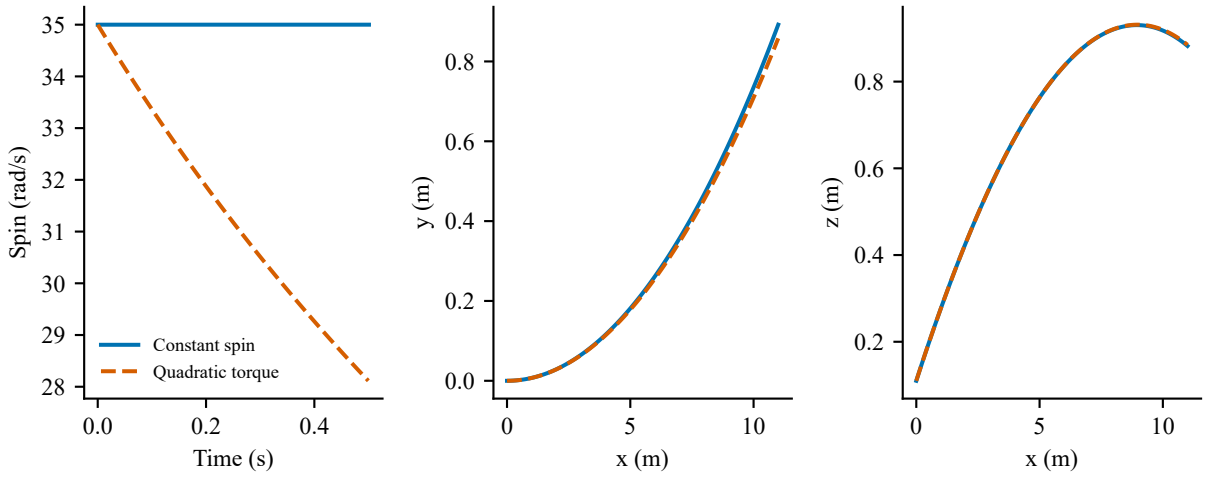



Figure 25: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for the prescribed curved shot. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped crossing is $(0.8579, 0.8848)$ m, compared with $(0.8921, 0.8822)$ m at constant spin. The displacement is 0.0343 m (0.312% of 11 m). Spin falls by 19.671%, and crossing time changes by -0.233%. Adding viscous torque shifts the damped crossing by 1.2e-05 m. The short-flight endpoint is less sensitive than spin magnitude to this reference torque.

## 6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [8]. At fixed coefficients, a wind speed $w$ along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{32}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 18 evaluates these rough estimates at the curved-shot launch speed. The listed wind speeds are illustrative conditions for comparison.

Table 18: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

| $w$ (m/s) | Headwind (%) | Tailwind (%) | Crosswind (%) | Angle (°) |
|---|---|---|---|---|
| 1 | +8.51 | -8.16 | +0.17 | 2.39 |
| 3 | +26.56 | -23.44 | +1.56 | 7.13 |
| 5 | +46.01 | -37.33 | +4.34 | 11.77 |

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

# 7 Conclusions

The model separates the mechanisms of gravity, drag and Magnus lift. Gravity produces downward curvature, drag reduces speed, and spin redirects the path. The complete Teamgeist relations connect coefficient changes to speed and rotation; the matched wind-tunnel comparison reproduces drag more closely than lift.

Five prescribed kick situations provide six initial states and 24 forward flights. The matched top- and backspin pair demonstrates the reversal of vertical Magnus force, while sidespin produces horizontal curvature. In the curved M4 example the goal crossing is $(0.8921, 0.8822)$ m at $0.5008$ s. The same initial state is used to quantify angular damping, giving a crossing displacement of 0.0343 m under the cited reference setting.

Three actual penalties supply distinct early sampling sequences and held-out trajectories. Coordinate conversion places all three in the competition geometry. The 12 inverse outcomes list fitted initial states, goal crossings, crossing deviations and ground termination separately, connecting the recorded motion to each force assumption. For the competition-position example in Section 5, a one-degree launch-direction change moves the crossing by about 0.20 m. With the early positions fixed, the explored compatible states shift the crossing by up to 0.1183 m; joint perturbations are quantified by 100,000 Monte Carlo trajectories with fixed ball mass and radius.

# 8 Model Evaluation

## 8.1 Physical Interpretation and Reproduction

The four-model hierarchy makes the role of each force visible under shared initial conditions. Forward and inverse calculations use the same state derivative and coefficient definitions. Original source coordinates, selected frames, transformed coordinates, numerical settings and candidate solutions are saved with the results.

## 8.2 Prediction and Further Development

The real-penalty comparison evaluates how early motion carries into the later path. Bound-active solutions and residuals identify the influence of the chosen coefficient scale and initial-state search. Ball-specific coefficients, direct spin measurements and higher-rate synchronized tracking would improve these connections. Rotational damping and wind can be included when their effect is relevant to the flight considered.

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# Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
--
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
        0.])
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```